

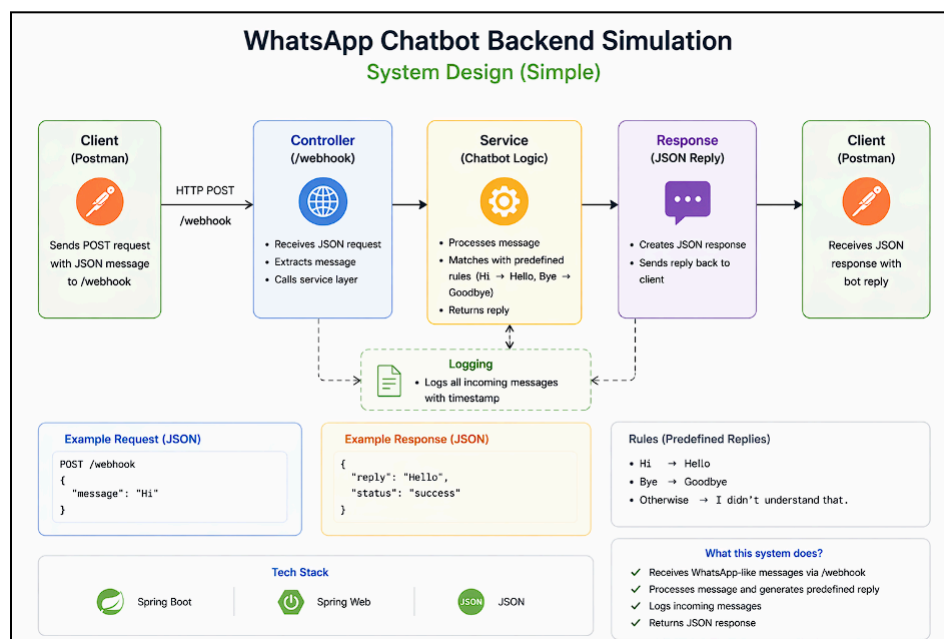
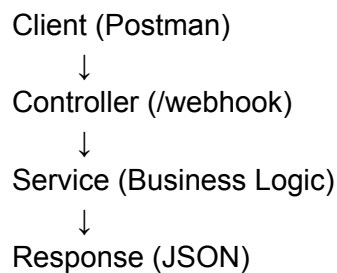
System design:

This project simulates a WhatsApp chatbot backend system using Java and Spring Boot. It demonstrates how a backend service receives, processes, and responds to incoming messages via a REST API.

Instead of forwarding messages to another user, the system generates automated responses based on predefined rules.

Architecture

The system follows a layered architecture:



Components

1. Client (Postman)

- Simulates a user sending messages
- Sends HTTP POST requests to the backend

2. Controller Layer

- Exposes REST endpoint: /webhook
- Accepts JSON input
- Validates and forwards request to service layer

3. Service Layer

- Contains core chatbot logic
- Processes incoming messages
- Matches predefined rules:
 - Hi → Hello
 - Bye → Goodbye
 - Default → fallback response

4. Response Handling

- Generates structured JSON response
- Returns reply to client

5. Logging

- Logs all incoming messages using SLF4J
- Helps in monitoring and debugging

◆ Data Flow

1. Client sends POST request to /webhook
2. Controller receives and parses JSON
3. Controller calls Service layer
4. Service processes message and determines reply
5. Response is generated
6. Controller returns response to client

♦ **API Design**

Endpoint

POST /webhook

Request Format

```
{  
  "message": "Hi"  
}
```

Response Format

```
{  
  "reply": "Hello",  
  "status": "success"  
}
```

♦ **Design Decisions**

- Layered Architecture for separation of concerns
- Rule-based logic for simplicity and clarity
- Stateless API design (no database required)
- JSON-based communication for standardization

♦ **Scope & Limitations**

Included:

- Message processing
- Predefined responses
- Logging

Not Included:

- Real WhatsApp API integration
- Real-time messaging (WebSockets)
- Database storage
- Authentication & security

◆ **Future Enhancements**

- Add NLP-based response handling
- Integrate database for message history
- Deploy to cloud platforms (Render)
- Extend to multi-user chat simulation

◆ **Conclusion**

This system demonstrates a simplified backend architecture of a messaging application, focusing on request handling, message processing, and response generation using Spring Boot.